Knowing When to Hedge:
Where Epistemic Calibration Comes From in Multi-Agent Pipelines

Sam Bobo

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Abstract

Epistemic calibration in a multi-agent LLM pipeline—qualifying claims where qualification is warranted and not where it is not—comes from a conditional instruction to the final model, evaluated against what the upstream models actually flagged. It does not come from the models, from the topology, or from training. We separate two things usually measured together. *How much* a pipeline qualifies is nearly free: a single well-prompted model matches a full three-specialist council (0.50 against 0.57, overlapping intervals). *Whether it qualifies appropriately* is not free, and is where every arrangement but one fails. Given a question that warrants no qualification at all, the lone prompted model hedges anyway (0.15 $[0.08, 0.22]$) while the council does not (0.00 $[0.00, 0.00]$, disjoint), because the council's instruction can check whether anything was flagged and the lone model cannot.

We then eliminate the obvious alternatives across $\sim$1,300 pre-registered runs on consumer hardware. It is not the models: four architectures produce indistinguishable qualification under a neutral prompt, and a domain-specialised medical model is no more careful than a generic one. It is not the topology: three specialists merged plainly reach 0.16 against one model's 0.14, and two rounds of debate add nothing. It is not the volume of upstream signal: tuning a specialist to qualify more doubles what it writes and makes the finished answer *worse*, monotonically in the number of specialists treated. And it is not training: weights failed at both ends of the pipeline, under three optimizers, at triple the data, and under a reward computed on the pipeline's own output. What remains is a mechanism we can state precisely and test: instructions can refer to evidence that exists only at inference time, weights cannot, and isolating the instruction clause by clause shows each lifting exactly the behavior it names and no other.

RETRACTED — the central claim is refuted, not merely unsupported. This draft argues that a conditional instruction to the final model suppresses unwarranted qualification. A controlled test of that claim, registered in advance and run after the draft was written, refutes it. The trigger-free question the original result rested on is off-topic for the specialists, so the planner consulted nobody and the pipeline bypassed the conditional instruction entirely; the celebrated 0.00 measured a prompt that contains no preservation clauses. Repeating the measurement on three trigger-free questions the planner *does* engage with—routing and clause application verified on every run—gives the council 0.64 $[0.36, 0.95]$ against a lone prompted model's 0.31 $[0.21, 0.42]$. The registered threshold was an upper bound below 0.10. The council is not better calibrated than a single model; on this evidence it is worse. The paper's thesis cannot be repaired by rewording and is withdrawn. Sections 3 and 5, which concern volume and do not depend on a trigger-free measurement, stand.

Correction (August 2026), continued. The claim that the models share a common *intrinsic band* under a neutral prompt is weakened: for three of the four, 70–80% of runs contain no epistemic marker at all, so the matching densities reflect a floor rather than a shared level, and per-character normalization coincidentally equates a model that emits $3.3\times$ more markers in $3.7\times$ more text. The supportable statement is that these models produce *almost none* unprompted and cannot be distinguished there. The contrast that carries the argument—the same specialist model at zero markers in 80% of neutral runs versus 1.31 as an instructed pipeline seat—is unaffected, being within-model and spanning the floor.

1 Introduction

This paper reports one finding. In a multi-agent pipeline, epistemic calibration is produced by a conditional instruction to the final model, evaluated against what the upstream models flagged—and by nothing else we were able to find.

The finding is not obvious because two properties usually measured together come apart. The first is *volume*: how much a system qualifies its claims on questions that warrant it. Volume turns out to be nearly free. One model with a paragraph of instruction reaches 0.50 qualifying behaviors per thousand characters; an entire council of three domain specialists behind a planner reaches 0.57, and the intervals overlap. Anyone measuring only this would conclude that orchestration is not worth its cost, and a growing literature does exactly that.

The second property is *discrimination*: whether the system stays quiet when nothing warrants qualification. Here the same two arrangements are not alike at all. On a question we wrote to warrant no hedging—every figure it needs is stated, nothing in it has changed in a decade—the lone prompted model hedges at 0.15 $[0.08, 0.22]$, opening with a section headed "Current State Snapshot (as of my training data)." The council hedges at 0.00 $[0.00, 0.00]$, on every run. The intervals are disjoint.

That gap is the paper. It exists because the council's instruction is *conditional*—if a specialist flagged an assumption, label it; if a specialist flagged that information may have aged, carry that through—and because there are specialists whose output the condition can be checked against. The lone model is given the same requirements as standing instructions, since it has no upstream to refer to, and so they fire whether or not anything warrants them. It has the form of epistemic care without the judgement.

Three assumptions do not survive this. That specialist models bring better epistemic judgement to their domains: a clinical model asked a question under a neutral prompt qualifies no more than a generic model of similar size. That more agents or more deliberation improve epistemic quality: three specialists merged plainly match one model working alone, and two rounds of debate match the merge. And that the behavior can be trained in: it could not be, at either end of the pipeline, by any method we tried.

We state the result first (§2), eliminate the alternatives (§3), give the mechanism (§4), and confirm it by taking the instruction apart clause by clause (§5).

Setup. Four models on one 32 GB machine: a planner/aggregator and three domain specialists (healthcare, legal, finance), all open-weights and quantised. The planner decomposes a question and sends each specialist a self-contained sub-question; the aggregator writes the final answer under the instruction reproduced in Appendix A. We count four families of epistemic behavior per thousand characters—disclosing that information may post-date training, labelling a number as an assumption, distinguishing jurisdictions, and calibrated hedging—by pattern matching, cross-checked on every load-bearing comparison by a calibrated entailment model and by blinded pairwise LLM judges (Appendix B). Seven questions, five seeds, bootstrap intervals; predictions registered before the runs that test them, including the ones that failed.

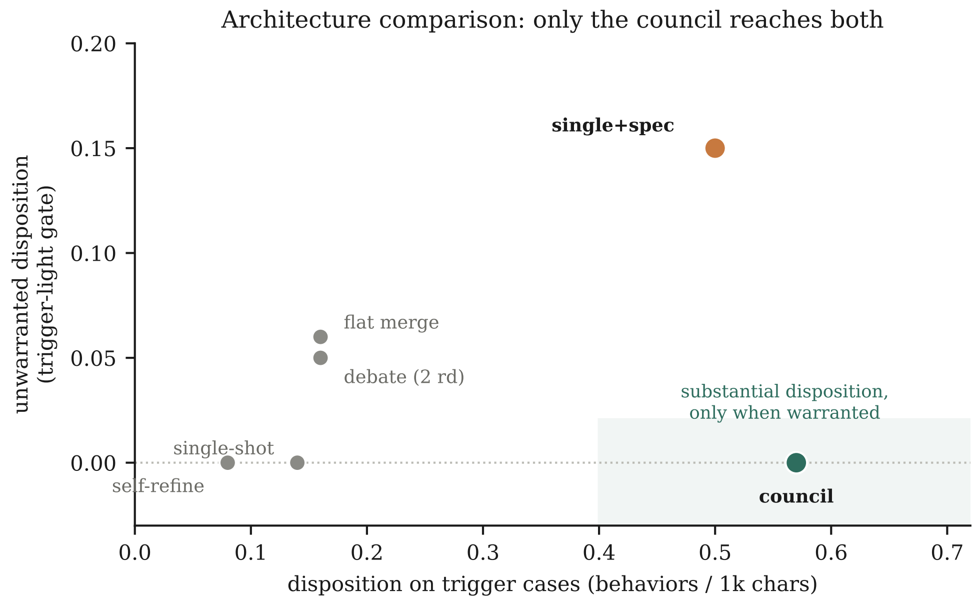

Figure 1: Six arrangements, one model in every role (210 runs). Horizontal: qualification where warranted. Vertical: qualification where it is *not*. Only the council reaches the shaded region.

Table 1: Volume and discrimination are different properties. Bootstrap 95% intervals over $n = 30$ (volume) and $n = 5$ (discrimination) runs per arm.

Arrangement	instruction	Volume	Unwarranted
single model	none	0.14 [0.09, 0.19]	0.00 [0.00, 0.00]
three specialists, plain merge	none	0.16 [0.11, 0.22]	0.06 [0.02, 0.11]
three specialists, two-round debate	none	0.16 [0.12, 0.20]	0.05 [0.00, 0.10]
single model, self-critique	none	0.08 [0.05, 0.12]	0.00 [0.00, 0.00]
single model	standing	0.50 [0.41, 0.60]	0.15 [0.08, 0.22]
three specialists, council	**conditional**	**0.57** [0.45, 0.71]	**0.00** [0.00, 0.00]

2 The result

To separate what the arrangement contributes from what the models contribute, we held the model fixed—the same 20B model in every role—and varied only the shape of the pipeline across six designs. Any difference is therefore attributable to structure and instructions alone.

Table 1 contains the finding. Read the volume column alone and the council is unremarkable: it beats the arrangements with no disposition instruction by three to four times, and ties the one arrangement that has an instruction. Read both columns and only one row does what a careful analyst does. The council is the sole arrangement combining substantial qualification with none where none is warranted.

The two ingredients are separable and neither suffices. Specialists without a conditional instruction (rows two and three) produce single-model volume: three models deliberating, even for two full rounds, add essentially nothing. An instruction without specialists (row five) produces the volume but fires indiscriminately, because there is nothing for it to condition on. Only their conjunction produces both.

3 What it is not

Four explanations are more natural than ours. We tested each.

3.1 Not the models

The most common justification for specialist pipelines is that domain models bring better judgement to their domains. This is checkable in isolation: take the specialists out, give them a neutral instruction, and see whether they qualify more than a generic model.

They do not. Med42-8B—a Llama-3 model tuned on clinical data and preference-aligned—produces 0.15 $[0.04, 0.28]$. Mistral-7B-Instruct produces 0.14 $[0.04, 0.26]$, Qwen2.5-7B-Instruct 0.14 $[0.06, 0.24]$, and a 20B mixture-of-experts 0.14. Four models across four architectures sit within 0.01 of one another.

The same clinical model, placed in the pipeline as a specialist, produces 1.31 $[1.02, 1.63]$—nearly nine times more, intervals nowhere near overlapping. Nothing about the model changed. What changed is that its in-pipeline system prompt says *"Flags training-cutoff uncertainty EXPLICITLY when guidelines or evidence may have evolved,"* and that the planner appends a further reminder when it judges the question time-sensitive—which it did in 30 of 34 runs. The specialist's epistemic richness is an instruction, not an expertise.

3.2 Not the topology

If not the models individually, perhaps the arrangement of them. Rows two and three of Table 1 rule this out. Three specialists whose answers are merged under a plain instruction reach 0.16, against a single model's 0.14. Two full rounds of debate, in which each specialist revises after reading the others, reach 0.16: more deliberation, no more care. A single model asked to draft, critique itself and revise reaches 0.08—the lowest of any arrangement, because self-criticism tightens prose and the qualifications go first.

3.3 Not the volume of upstream signal

Perhaps the specialists simply need to send more. This is the alternative we expected to be true, and it fails in the opposite direction from a null.

Prompting the legal specialist to qualify more than doubles what it writes, from 0.84 to 1.82; fine-tuning it on qualification-rich examples does the same, 1.75. Both changes are real—the intervals exclude the untrained baseline and the text is visibly more careful. Both make the finished answer *worse* than leaving the specialist alone: 0.92 against an untrained 1.21. Extending the treatment across specialists confirms the direction: zero, one, two and three treated specialists give final answers of 1.01, 1.03, 0.84, 0.64, a monotone decline ($\rho = -0.80$).

Qualification does not accumulate upstream and get diluted. It is not accumulating at all. A specialist told to flag uncertainty everywhere flags it in every clause, and the aggregator—whose instruction asks for a clean integrated answer—reads that as noise and edits it out.

3.4 Not training

The remaining alternative is that we instructed where we should have trained. We trained at both ends of the pipeline, by three methods, under predictions registered in advance.

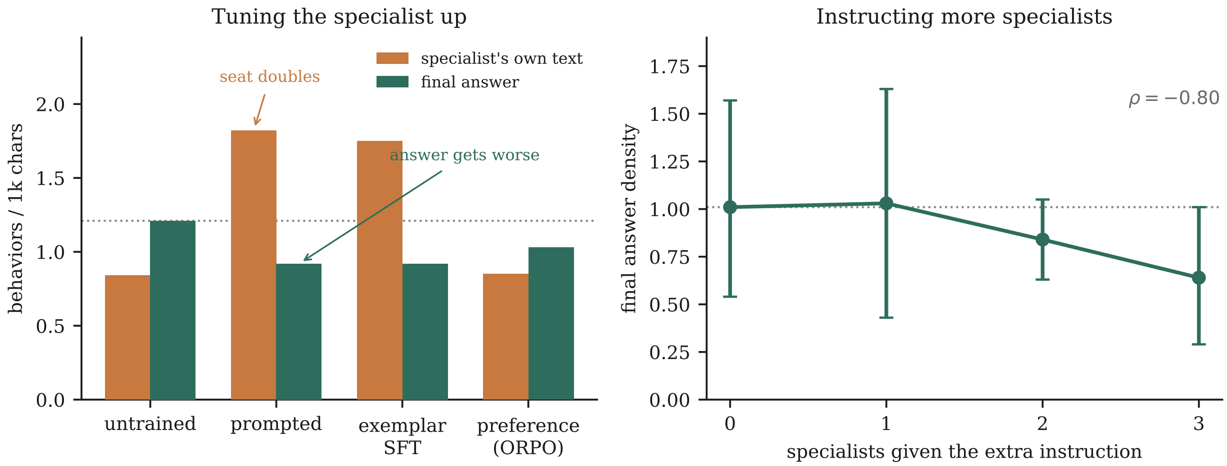

Figure 2: *Left:* treatments that raise what a specialist writes lower what the pipeline finally says; dotted line is the untrained final answer. *Right:* extending the treatment from zero to three specialists lowers the final answer monotonically.

At the specialist, preference optimization installs no additional qualification: 0.85 against an untrained 0.84. Tripling the training data changed nothing, even though the larger corpus raised the model's training-time accuracy at telling good examples from bad from 0.48 to 0.94—the preference was learned and not acted on. Replication on a second specialist of a different lineage produced no effect, and on a third an effect we had registered in advance failed to appear.

At the aggregator—the position this paper says decides everything—training also fails. Fine-tuning it on qualification-rich synthesis examples left its output where it started, 0.89 $[0.67, 1.11]$ against 1.03 $[0.79, 1.28]$ untrained. We repeated this with the reward corrected in every way we could think of: sampling the aggregator's own outputs rather than a fixed corpus, and scoring each on the finished answer for qualifying heavily where warranted and not at all where not. It moved nothing: 0.60 $[0.41, 0.82]$ against 0.56 $[0.37, 0.77]$.

4 The mechanism

Something has to explain why an instruction does what four other interventions could not. We think three things do, and we can test two of them.

4.1 Conditionality, and why it needs witnesses

Every clause of the aggregator's instruction that does any work has the same form: *if* a specialist did something, *then* carry it through. The trigger is not a standing property of the domain but a fact about this particular run—whether this specialist, on this question, flagged an assumption or a stale figure.

This is why the specialists matter, and it is a narrower role than the one usually claimed for them. They are not there to produce more qualification; the aggregator's instruction sets that, and §3 showed the specialists barely affect it. They are there to produce a *record of what was uncertain*, against which a conditional can be evaluated. Their value is that they can be checked.

That also explains the failure mode of the lone prompted model. Given the same requirements with no upstream to refer to, a conditional instruction has to be flattened into a standing one—*disclose training-cutoff uncertainty* rather than *if a specialist flagged it, disclose it*—and a standing instruction fires whether or not the occasion calls for it. The 0.15 is not a failure of the model's judgement. It is the arithmetic consequence of asking for a conditional behavior without supplying anything to condition on.

It also explains why treating specialists as amplifiers backfires (§3). A witness who says everything is uncertain carries no information; the conditional fires everywhere and discrimination collapses, which is exactly what the monotone decline shows.

4.2 Why instructions and not weights

Three arguments, in increasing order of what they cost us to establish.

First, the behavior is not absent from the weights—it is present and unused. Four models produce 0.14 under a neutral prompt and reach 0.5–1.2 when instructed. Nothing was installed in between. An instruction is therefore asking for something the model already does; training is trying to change what it does by default. Those are different operations, and only the first is cheap.

Second, the target behavior is conditional on information that does not exist until inference. Weights encode a disposition a model brings to every input. They cannot refer to a particular thing that arrived in this context; an instruction can, because it points at the context. This is not a claim that models cannot learn context-dependent behavior in general—plainly they can—but that learning *this* condition would require the model to discover a pattern it is not currently attending to, where an instruction merely names it.

Third, and most directly: training on the behavior did not consume the instruction's effect. After fine-tuning the aggregator on precisely the behavior its instructions elicit, the instructions still worked at full strength—adding clauses produced a $1.58\times$ rise on the trained model against $1.82\times$ on the untrained one. If weights and instructions were competing routes to the same behavior, training on that behavior should have absorbed part of the instruction's job. It did not, which suggests they act on different things: the weights on what a model tends toward, the instruction on what it does this time.

What we cannot claim. Our training was small—low-rank adapters over tens of preference pairs, against instruction-following that model builders installed with orders of magnitude more data. "A small amount of training did not overwrite a deeply installed capability" is a much weaker statement than "weights cannot do this," and it is the one our data supports. What holds without qualification is that at every scale available to us the instruction was free, immediate and effective, and the training was none of those.

5 Isolating the instruction

If the mechanism is right, the instruction should behave like a specification rather than like general encouragement—each clause should move the behavior it names and leave the others alone. We tested this by isolating clauses against a baseline with none, one final model, everything else fixed (210 runs).

Three things follow. Each clause moves its own named behavior and nothing else—numeric framing raises labelled assumptions from 0.003 to 0.499 and leaves the rest flat; the caveats clause raises cutoff disclosure and leaves the rest flat. These are targeted specifications, not general priming, which rules out the alternative that instructions simply put the model in a more cautious mood.

Second, the composite gain goes to whichever single clause names the behavior that this model already produces in quantity. Under this aggregator that is numeric framing, which alone accounts for the whole

Table 2: One clause at a time. Each clause lifts the behavior it names and no other; every arrangement holds the trigger-free control at zero.

Clauses present	Volume	Named behavior	Unwarranted
none	0.20 $[0.14, 0.26]$	—	0.00
tension acknowledgment only	0.19 $[0.11, 0.27]$	(names none)	0.00
numeric framing only	$\mathbf{0.64}$ $[0.53, 0.77]$	$0.003 \to 0.499$	0.00
vocabulary only	0.18 $[0.14, 0.22]$	$0.000 \to 0.014$	0.00
caveats only	0.20 $[0.14, 0.26]$	$0.000 \to 0.035$	0.00
all four	0.55 $[0.46, 0.67]$	—	0.00

effect; the caveats clause raises its own behavior correctly but from a base too small to move the total. Under an aggregator with a different profile we would expect a different clause to carry it, and our ledger shows the profile does vary by model. This is a scoping condition on any practical recommendation: write the clause for the behavior your writer actually produces.

Third, and most relevant to the mechanism: every arrangement, including each single clause, holds the trigger-free control at exactly zero. Conditionality is a property of each clause separately, not something that emerges from assembling them—each is an if-then over upstream evidence, so none fires when nothing was flagged.

6 Implications

The practical consequences are narrow and, we think, actionable.

Measure both properties. A pipeline evaluated only on how much it qualifies will look indistinguishable from a well-prompted single model, and the comparison will conclude that orchestration is not worth its cost. Every evaluation of epistemic behavior needs at least one case that warrants none.

Choose the final model for instruction-following, not for disposition. Models differ far less in what they produce unprompted than in how strongly they answer an instruction. We encountered one model that could not follow instructions in any message position and could not emit the planner's structured output; whatever its latent qualities it was unusable in this role.

Write the aggregator's instruction conditionally. Not *be careful*, but *if a specialist flagged X, carry X through*. This is the whole difference between the last two rows of Table 1.

Build specialists to flag accurately, not abundantly. Tuning them to qualify more degrades the finished answer. Their job is to be checkable.

Do not spend on fine-tuning components for epistemic behavior. At both ends, three optimizers, triple the data, and a corrected reward, it never reached the page.

7 Limitations

Of five behaviors we set out to count, one fell below detection on both instruments, so we report four; the effect of instructions is clear in combination but not for any single behavior alone. Pattern matching misses paraphrase and rewards brevity; two further instruments agree with it on the load-bearing comparisons, but all three are automatic and none is anchored to human judgement.

The headline rests on the trigger-free control, and that control is currently one question at five runs per arm. It is the single thinnest support in the paper and the first thing we would expect a reader to press on; an expansion to additional trigger-free questions is registered and pending.

Everything ran at or below 20B parameters on one machine with five seeds and seven questions we wrote ourselves; a frontier aggregator could behave differently. The architecture comparison uses one model family in every role and one implementation of each design, with debate and self-critique in their simplest standard forms. The clause isolation ran on a single aggregator whose behavior profile we later found unusual, so its dependence on the model is inferred rather than tested.

We report defects in our own record rather than repairing them. One model intended as a comparison produced degenerate output—26 of 30 answers were fragments ending mid-sentence—which we found only after computing a figure from them; that arm is withdrawn, and with it a lineage comparison we had intended to make. In our earliest arms, five runs per question are one run from an earlier session plus four from a later one, and 133 runs record a batch label where a model identifier belongs. None of this changes a reported interval.

8 Conclusion

Epistemic calibration in these pipelines is not a property of the models, of how many of them there are, of how much they say, or of how they were trained. It is produced by an instruction to the final model that refers to what the earlier models flagged, and it disappears when either half of that is missing. A lone model given the same instruction qualifies just as much and qualifies indiscriminately, because it has nothing to refer to. A council whose specialists are trained to qualify everything degrades, because a witness who always says the same thing carries no information.

The reason an instruction succeeds where training failed appears to be that the behavior wanted is not a disposition at all. It is a decision about the present occasion, conditioned on evidence that exists only once the earlier models have spoken. Weights set what a model tends to do; only an instruction can point at what just arrived.

Reproducibility. Code, prompts, all runs, pre-registrations, protocol amendments, refuted predictions, and an integrity audit that withdrew one arm: `https://github.com/srbobo/council-of-experts-research`.

A The aggregator's instruction

The aggregator works in two steps—identify tensions between the specialists, then write the integrated answer. The second step carries five requirements, of which three are the preservation clauses. Verbatim:

> **1.** Acknowledge the tensions you just identified—not abstractly, but at the points where they bite the answer.
>
> **2. PRESERVE numeric framing**—if a specialist labeled a number as a modeled assumption, your synthesis MUST also label it as an assumption (use "modeled at," "assumed"). Never adopt a specialist's modeled number as a fact.
>
> **3. PRESERVE precise vocabulary**—if a specialist used a specific legal term, jurisdiction, FDA pathway, or clinical concept, preserve it; do not smooth out precision in the name of accessibility.

4. **PRESERVE caveats**—if a specialist flagged training-cutoff uncertainty or data fragility, propagate that flag into your synthesis.

5. Use whatever structure best fits the user's question.

Clauses 2–4 are conditional on something a specialist did. Clause 1 is not, and when tested alone it changes nothing—which is the cleanest available demonstration that conditionality, rather than the presence of an extra instruction, is what does the work.

B Instruments

Pattern matching over four behavior families is the primary measure. Two independent instruments check every load-bearing comparison: an entailment model calibrated against construction-labelled pairs, and blinded pairwise LLM judges shown both texts in each order and required to quote the evidence for their verdict. Where all three agree we report the result; where they disagree we report the disagreement. One finding did not survive—an apparent *increase* in hedging on the finance specialist appeared under pattern matching alone, and neither other instrument saw it. We retired it as an artifact of the measure.